ZADANIE A4

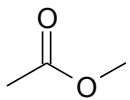
Reakcje związków organicznych

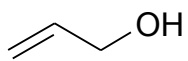
Polecenia:

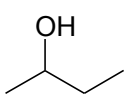
a. W podanych niżej wzorach związków A – G wskaż i nazwij grupy funkcyjne.

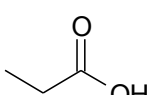
b. Podaj wzory głównych produktów wszystkich reakcji.

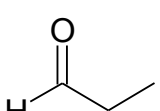
c. Wskaż, które z podanych substratów są izomerami i które z nich zawierają centrum stereogeniczne (asymetryczny atom węgla).

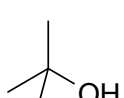
A.  + NaOH ⟶

B.  + H_2O $\xrightarrow{H^+}$

C.  + $KMnO_4$ ⟶

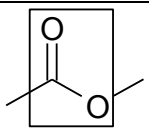
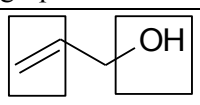
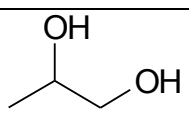
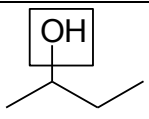
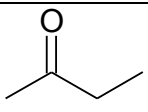
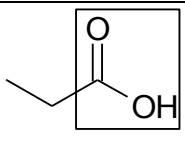
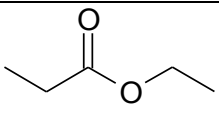
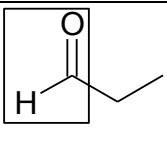
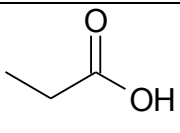
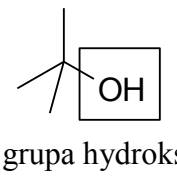
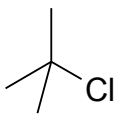
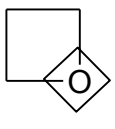
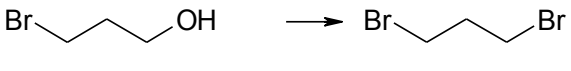
D.  + C_2H_5OH $\xrightarrow{H^+}$

E.  + $Cu(OH)_2$ ⟶

F.  + HCl ⟶

G.  + HBr ⟶ $\xrightarrow{HBr}$

ROZWIĄZANIE ZADANIA A4

	Polecenie ***a.***		Polecenie ***b.***
A.	 grupa estrowa	+ $NaOH$ ⟶	CH_3COONa + CH_3OH
B.	 podwójne wiązanie, grupa hydroksylowa	+ H_2O $\xrightarrow{H^+}$	
C.	 grupa hydroksylowa (alkohol drugorzędowy)	+ $KMnO_4$ ⟶	
D.	 grupa karboksylowa	+ C_2H_5OH $\xrightarrow{H^+}$	
E.	 grupa aldehydowa	+ $Cu(OH)_2$ ⟶	
F.	 grupa hydroksylowa (alkohol trzeciorzędowy)	+ HCl ⟶	
G.	 grupa eterowa (eter cykliczny)	+ HBr ⟶	

Polecenie ***c.*** Izomerami są: A i D ($C_3H_6O_2$), B, E i G (C_3H_6O), C i F ($C_4H_{10}O$).
Asymetryczny atom węgla występuje w substracie C.